Team_10537 at the NTCIR-19 AI CUP-VeriPromiseESG Task: Hierarchical Projection as a Validity Layer for Multilingual ESG Promise Verification

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ABSTRACT

ESG promise verification requires jointly predicted fields to satisfy hard parent–child constraints, particularly when outputs are intended for audit-oriented use. We study hierarchical projection as a retraining-free validity layer for field-wise classifiers. Seven decision rules are compared on identical cross-fitted probabilities and test rows in Chinese, followed by projection-versus-argmax replication across English, French, Japanese, and Korean corpora. Projection reduces the Chinese invalid-tuple rate from 12.55% to zero by construction. Its official weighted macro-F1 contrast is undetected ($\Delta = -0.001$, $p_{\text{Holm}} = 1.000$), while pre-specified tuple accuracy improves by +0.035 ($p_{\text{Holm}} = .001$). Across 32 external backbone-by-loss arms, tuple accuracy improves in all 32, whereas weighted macro-F1 improves in 14. These results position hierarchical projection not as a guaranteed route to higher field-wise scores, but as a practical way to guarantee coherent outputs and repeatedly improve complete-output correctness, with field-score trade-offs varying across models and corpora. Validity alone does not settle which structured rule to use: searching all 17 legal states instead of projecting greedily costs −0.006 tuple accuracy ($p_{\text{Holm}} = .028$), an adverse pre-specified result we report in full. Code and reproducibility artifacts are available at https://github.com/tom1030507/ntcir19-esg-validity-layer.

KEYWORDS

ESG, promise verification, hierarchical classification, constrained decoding, evaluation metrics

TEAM NAME

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SUBTASKS

AI CUP Special Session at NTCIR

1 INTRODUCTION

We participated in the VeriPromiseESG task of AI CUP 2026 [1, 8], Taiwan's national inter-collegiate artificial intelligence competition, held as the AI CUP special session of NTCIR-19. The task definition, corpus, and official evaluation are the organizers' and are not restated here. All results below are computed on the labeled development release, because the competition test labels are not available to participants.

Environmental, social and governance (ESG) promise verification is intended to support audit-oriented decisions, for which internal consistency is a basic requirement. Promise status, verification timeline, evidence status, and evidence quality describe one claim, and their parent–child implications determine whether the combined output is usable as a coherent record. A paragraph predicted to contain no promise cannot coherently receive a substantive timeline, and a prediction of no evidence cannot coherently receive an evidence quality. Individually plausible field labels can therefore form an invalid tuple.

Field-wise weighted macro-F1 is a standard measure of classification quality for multi-field tasks, and it is the official ranking metric here. It does not, however, assess whether the four predictions form a jointly legal tuple. A validity-preserving rule may therefore repair an unsupported child without receiving a corresponding field-score gain. The metric places no weight on legality: because each field is scored on its own, a correct field still earns its weight even when the ancestors that would license it are wrong, so an invalid tuple retains 44.3% of the available weighted field credit on average. Table 5 derives that figure term by term. We ask whether a retraining-free inference constraint can guarantee legal outputs, and how its effects differ under field-wise and whole-tuple evaluation.

We answer it with a controlled decision comparison, not a model or competition-performance claim: seven rules consume identical cross-fitted probabilities and test rows, and projection guarantees a legal tuple without retraining the classifier. In Chinese the confirmatory official-score family is null, yet pre-specified whole-tuple accuracy improves; across 32 external arms in four ML-Promise languages, tuple accuracy improves in all while weighted macro-F1 has a mixed sign. These are within-corpus contrasts; raw scores do not isolate language effects.

Our contributions are:

(1) We formalize the supplied hierarchy as 17 legal tuples out of 120 and verify that all 2,000 audited Chinese paragraphs from 49 companies are legal.

(2) We evaluate projection as a retraining-free validity layer in a seven-rule controlled comparison, reporting the complete confirmatory official-score family and the pre-specified tuple family, including adverse results.
(3) We externally replicate the within-arm projection contrast across four languages and all 32 external arms, reported one per row in Table 7, while keeping post-hoc metrics and structural and backbone screens explicitly exploratory.

2 RELATED WORK

The AI CUP task uses the four-part formulation described in the PromiseEval shared-task overview: identify an ESG promise, determine whether it has actionable evidence, assess the clarity of the promise–evidence pair, and assign a verification horizon [2]. ML-Promise is the associated multilingual corpus resource; we use its English, French, Japanese, and Korean releases for external replication [16]. Unlike ClimateBERT-NetZero, which detects net-zero and reduction targets [15], the task jointly assesses promise status, evidence, its quality, and timing.

These dependent outputs form a compact instance of hierarchical text classification, in which labels occupy a supplied structure and coherent predictions respect its ancestor relations [14]. The AI CUP hierarchy comes from the task's official field logic: we neither discover the hierarchy nor propose a new hierarchical architecture. Instead, we compare decision rules over the same field-wise probabilities while preserving or ignoring that given structure.

Yu et al. introduced path-constrained MicroF1 and MacroF1, under which a correct node receives credit only when all its ancestors are also correct [18]. Ji et al. later named this family *C-metrics* [12]. Our path-constrained weighted macro-F1 (C-wF1) adapts Yu et al.'s principle to this task's fields and its official weighted macro-F1; it is therefore not the original C-MacroF1. Section 5.2 defines both it and the set-overlap hierarchical F1 (hF). Both were adopted post hoc and are reported as exploratory.

Plaud et al. observed from point estimates that constrained F1 variants did not globally change method rankings [14]. Ranking a collection of models is a different estimand from our paired difference between decision rules on shared rows and probabilities. Of the two post-hoc metrics, only C-wF1 isolates the treatment of children whose predicted ancestors do not support them while keeping the official field aggregation.

3 TASK AND DATA

Outputs and legal states. The task predicts four fields for each paragraph. We study the AI CUP 2026 VeriPromiseESG development release [1]. Its label inventory and parent–child logic follow PromiseEval [2]. Promise status (PS) is *Yes* or *No*. Verification timeline (VT) is *already*, *within two years*, *between two and five years*, *more than five years*, or *N/A*. Evidence status (ES) is *Yes*, *No*, or *N/A*; evidence quality (EQ) is *Clear*, *Not Clear*, *Misleading*, or *N/A*. These fields obey hard implications: PS=No requires VT=ES=EQ=N/A, whereas PS=Yes requires one of the four substantive timelines and ES in {Yes, No}. In turn, ES=No requires EQ=N/A, whereas ES=Yes requires one of the three substantive quality labels. Thus the Cartesian label space has $2 \times 5 \times 3 \times 4 = 120$ combinations, but only

$$1 + 4(1 + 3) = 17 \tag{1}$$

Table 1: AI CUP 2026 dataset composition and hierarchy audit.

Statistic	Development	Competition test
Paragraphs	2000	2000
Source reports (PDFs)	49	49
shared with other split	49	49
Companies	49	49
Gold labels available	Yes	No
Gold hierarchy-valid tuples	2000 / 2000	n/a
Legal states observed	15 / 17	n/a
Rarest classes		
within_2_years	34	n/a
Misleading	2	n/a

legal tuples: one for PS=No, and for each of four timelines either one ES=No state or three ES=Yes quality states. We write $\mathcal{S}$ for this set of 17 legal states throughout. Two terms and no others are used for the boundary it draws: a state is *legal* when it belongs to $\mathcal{S}$, and an emitted tuple is *invalid* when it does not.

Development-set audit. The release contains 2,000 labeled development paragraphs from 49 source reports and 49 companies. All 2,000 development rows obey the hierarchy, collectively realizing 15 of the 17 legal states (Table 1). The rarest EQ label, *Misleading*, has only two instances; we therefore make no improvement or significance claim for that class.

All 49 development reports also occur in the unlabeled competition test data. Because competition test labels are unavailable, we infer no label-derived test statistic. Each company contributes one report. Consequently, in this corpus a document-disjoint split is also company-disjoint, rather than a separate company-level experiment.

External replication corpora. ML-Promise provides labels in the same four fields [16]. After normalizing its releases to the common 17-state vocabulary, we retain 400 English paragraphs from 9 reports, 400 French paragraphs from 9 reports, and 400 Japanese paragraphs from 19 reports. The Korean release supplies labels, report URLs, and page numbers but no text; we extract the corresponding 500 PDF pages from 32 reports and use the first 384 tokens of each page. The extracted Korean text is not redistributed. Korean has no gold *Misleading* example, and its page-level input is not directly comparable to the paragraph inputs. We therefore interpret only within-corpus paired decision contrasts, never raw score differences between languages.

4 METHODS

4.1 Shared Base Models

For the Chinese study, we use `hfl/chinese-roberta-wwm-ext-large` as the fixed encoder, where `hfl` is the Joint Laboratory of HIT and iFLYTEK Research. The checkpoint has BERT-Large dimensions, 24 layers and hidden size 1024 [9]. Cui et al. document the checkpoint's Chinese RoBERTa-WWM family: it uses whole-word masking, extended pretraining data, and no next-sentence prediction [7].

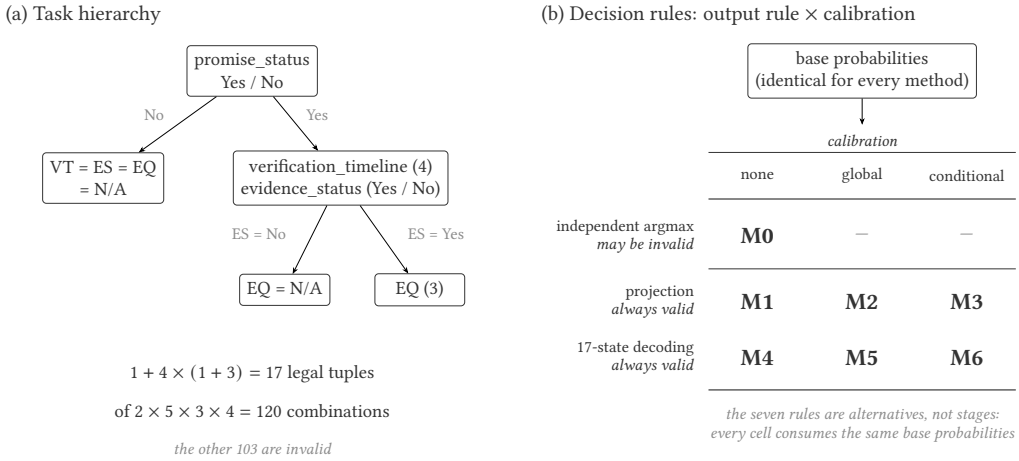


Figure 1: The task hierarchy and seven alternative decision rules, which share base probabilities but differ in calibration and validity guarantees.

We mean-pool its token representations and attach four independent linear heads, one per field. The encoder is shared, but neither it nor the heads condition on predicted ancestors. The input is paragraph text only; report URL, company, ticker, and page metadata are excluded.

All fits use the same frozen fine-tuning recipe: maximum length 384, 12 epochs, minibatches of 8 with two-step gradient accumulation (effective size 16), and AdamW with backbone and head learning rates 2×10^{-5} and 10^{-4}. The schedule is cosine with 10% warm-up and layer-wise decay 0.9. The four cross-entropy losses use the official PS/VT/ES/EQ weights 0.20/0.15/0.30/0.35 and label smoothing 0.1. For Train size n, K_t classes present in field t, and class support $n_{t,c}$, the class weight is $\min\{10, n/(K_t\, n_{t,c})\}$ when $n_{t,c} \geq 5$, and 1 otherwise. Hierarchy masking is disabled. We average the last three epoch checkpoints and perform no label-driven early stopping.

For the external fixed arms, English uses RoBERTa-large [13] and French, Japanese, and Korean use XLM-R-large [5]. English robustness runs add large DeBERTa-v3 [10] and ELECTRA [4] checkpoints plus RoBERTa-base. The other languages add RemBERT [3], XLM-R-base, and mBERT [9]. The Chinese anchor adds three exploratory screens of its own, distinct from those English checkpoints: a Chinese DeBERTa-v2-320M [19], a Chinese ELECTRA-large discriminator [6], and the base-sized checkpoint of the primary encoder's own family [7]. Every backbone uses the same maximum length, optimization recipe, three seeds, and five rotations. For each backbone we run the ordinary objective ($\lambda = 0$) and a fixed hierarchy-aware objective ($\lambda = 0.3$):

$$\mathcal{L} = \mathcal{L}_{\text{fields}} - \lambda \log \sum_{s \in \mathcal{S}} \prod_{t} p_{t,s_t}(x). \tag{2}$$

Here $\mathcal{L}_{\text{fields}}$ is the sum of the four weighted cross-entropy losses just described, and $\mathcal{S}$ is the set of 17 legal states of Eq. (1). The second term is the semantic loss of Xu et al. [17]: it penalizes probability mass outside $\mathcal{S}$ without choosing which legal state is correct. Structural-loss and alternate-backbone comparisons are exploratory; the primary decision comparison uses $\lambda = 0$.

4.2 Metric-Aware Decision Calibration

Calibration is the second factor of the comparison: the results table crosses three calibration settings with the two validity-preserving output rules, and each rotation's bias estimates are shared across both rules, which is what lets Table 2 be read as a factorial rather than as separate systems.

For field t and class c, decision calibration adds a class bias in log probability space:

$$z_{t,c}(x) = \log p_{t,c}(x) + b_{t,c}. \tag{3}$$

This changes decision boundaries and is not a claim of probability calibration. Each field's bias vector is fitted by deterministic coordinate ascent with that field's macro-F1 on the held-out Calibration partition as the objective, before either structured output rule is applied. The four objectives therefore decouple. Within every rotation, global and conditional biases are fitted separately; that rotation's global fit is shared by M2/M5 and its conditional fit by M3/M6.

Global biases use every Calibration row. Conditional biases use all rows for PS, only gold PS=Yes rows for VT and ES, and only gold ES=Yes rows for EQ. The three child N/A classes are structurally absent from these subsets. Their biases are therefore unidentifiable and fixed as

$$b_{\text{VT,N/A}} = b_{\text{ES,N/A}} = b_{\text{EQ,N/A}} = 0. \tag{4}$$

This has no effect on projection, which emits N/A only when its parent requires it. It does affect joint scores for (No, N/A, N/A, N/A); hence M6–M5 compares conditional with global estimation including their different estimable parameter sets.

Biases start at zero and are fitted by deterministic coordinate ascent over $[-3, 3]$ in increments of 0.05. Coordinates are visited in the order PS, VT, ES, EQ and, within a field, in the released label order (for EQ: Clear, Not Clear, Misleading, N/A); a move requires a strict improvement, ties favor the smaller absolute bias (then the

more negative value), and search stops after a pass without improvement or at 20 passes. Eligibility is determined from the corresponding Train subset: a class absent there is excluded from coordinate ascent and retains bias zero. Among Train-eligible classes, one absent from the matching Calibration subset also retains bias zero and is recorded as an absent-class fallback.

4.3 Validity-Preserving Output Rules

Complete projection. We treat projection as a validity layer over compatible field-wise probabilities, rather than as a new encoder or a claim of improved base-model accuracy. Projection is greedy and top-down. At each field it selects the highest-scoring class allowed by the chosen ancestors. Under PS=No, it sets VT, ES, and EQ to N/A. Under PS=Yes, it excludes N/A from VT and ES and chooses each field's best substantive class. Under ES=No, EQ becomes N/A; under ES=Yes, projection chooses EQ's best substantive class. These rules completely enforce both directions of every parent implication. This bidirectional projection always returns one of the 17 legal states; a one-way N/A overwrite would not.

17-state decoding. The alternative scores every $s \in \mathcal{S}$ and returns

$$\hat{\mathbf{y}} = \arg\max_{s \in \mathcal{S}} \sum_t \{\log p_{t,s_t}(x) + b_{t,s_t}\}. \tag{5}$$

The four fields enter this sum unweighted. A per-field scale α_t would be the obvious generalization, and we deliberately do not fit one: the comparison against projection is meant to isolate the choice of output rule, and a decoder carrying four tuned parameters that projection does not have would no longer answer that question.

Figure 1 summarizes seven *alternative* rules over identical cross-fitted base probabilities, not sequential stages: M0 is uncalibrated independent argmax; M1/M2/M3 use projection with respectively no, global, or conditional bias; and M4/M5/M6 use 17-state decoding with the same three bias choices. Only M0 can emit an invalid tuple.

5 EXPERIMENTAL DESIGN

We first define the shared evaluation protocols and inference. Table 2 then summarizes the controlled comparison interpreted in Section 6.

5.1 Cross-Fitted Evaluation Targets

We use five-way rotating three-way cross-fitting. For rotation k, Test is fold k, Calibration is fold $(k+1) \bmod 5$, and Train comprises the other three folds; the three partitions are mutually exclusive. After five rotations, every development paragraph has served as Test exactly once. We concatenate the 2,000 out-of-fold predictions and compute each metric once, rather than average fold-level F1 values whose present gold-label sets may differ. The three seeds 42, 123, and 456 each determine both a fold assignment and training randomness, so seed variation represents the complete pipeline.

The primary, document-disjoint protocol balances report row counts with a randomized greedy procedure. After a seeded shuffle, it processes reports from largest to smallest and randomly places each in one of the two currently lightest folds. It estimates unseen-report generalization and, given the corpus audit, is also company-disjoint.

The same-document protocol instead jointly stratifies by gold PS, ES, and VT. It shuffles rows within each joint-label bucket and deals them round-robin to the five folds. Every report represented in Calibration or Test must also occur in Train through a different paragraph. This estimates seen-report, unseen-paragraph generalization. The two protocols target distinct estimands; neither is a biased and the other an unbiased version of one quantity.

Within each protocol and seed, M0–M6 consume the same Test rows and base probabilities.

External runs use document-disjoint report grouping, three seeds, and five rotations. Each language has eight paired arms – four backbones under both $\lambda = 0$ and 0.3 – giving 32 external arms, none of them chosen on an observed score. The same-document protocol is descriptive and available only for the primary large backbone. The complete executed study contains 765 training artifacts and 1,050 row-level prediction files across five corpora.

5.2 Metrics and Inference

The field-wise weighted macro-F1 is a standard choice for multi-field classification, but it contains no term that assesses the joint validity of a four-field tuple. For Chinese, the primary metric is the official weighted macro-F1,

$$\text{wF1} = 0.20F_{\text{PS}} + 0.15F_{\text{VT}} + 0.30F_{\text{ES}} + 0.35F_{\text{EQ}}, \tag{6}$$

where each F_t is field-level macro-F1 over classes present in the gold labels being scored. Because this denominator depends on gold support, per-class leverage differs by corpus: a unit gain on one EQ class contributes $0.35/3 = 0.1167$ to the Korean weighted total but $0.35/4 = 0.0875$ elsewhere. Pre-specified whole-tuple accuracy requires all four fields to match. Path-constrained weighted macro-F1 (C-wF1) and hierarchical F1 (hF) were adopted post hoc. C-wF1 retains the official field weights but counts a substantive child prediction as wrong when its predicted ancestors do not support it. Following the hierarchical precision/recall/F1 set-overlap definition [14], our hF micro-averages over emitted non-N/A field–value nodes: hP is total gold–prediction overlap divided by predicted nodes, hR divides the same overlap by gold nodes, and hF is their harmonic mean. For an inconsistent emitted tuple, we do not add missing ancestors before scoring; legal M1–M6 are unaffected because the conventions coincide for them. hF changes both consistency treatment and macro-versus-micro averaging, so the two supplementary metrics answer different questions.

ML-Promise has no official weighted metric. We therefore label its scores "AI CUP weights applied to ML-Promise." All external comparisons remain paired within corpus, backbone, and λ.

For paired method contrasts, we perform a paired PDF-cluster bootstrap over 49 source-report clusters. Each of 10,000 replicates samples 49 reports with replacement using seed 20260814, applies the same sampled rows to both methods and all three seeds, computes the within-seed differences, and averages those differences. The five pre-specified contrasts are M1–M0, M3–M2, M4–M1, M6–M3, and M6–M5. M1–M0 changes only the output rule, with calibration held out of both arms; it is therefore the cleanest test of projection rather than the highest-scoring configuration. Our

Table 2: Document-disjoint cross-fitted results on the Chinese development set.

ID	Calibration	Decoding	PS	VT	ES	EQ	wF1 (official)	Tuple acc.	Invalid %
M0	None	Independent	0.796	0.464	0.650	0.424	0.572±0.003	0.359±0.007	12.55
M1	None	Projection	0.796	0.467	0.651	0.418	0.571±0.003	0.394±0.004	0.00
M2	Global	Projection	0.796	0.493	0.650	0.418	0.574±0.004	0.428±0.004	0.00
M3	Conditional	Projection	0.796	0.501	0.651	0.416	0.575±0.003	0.432±0.004	0.00
M4	None	17-state	0.799	0.468	0.648	0.420	0.571±0.005	0.388±0.003	0.00
M5	Global	17-state	0.796	0.493	0.649	0.423	0.576±0.001	0.430±0.001	0.00
M6	Conditional	17-state	0.784	0.494	0.634	0.418	0.567±0.008	0.427±0.005	0.00

Table 3: Chinese paired contrasts on the two pre-specified metrics, with uncorrected 95% report-cluster bootstrap intervals and Holm-adjusted p-values.

Metric	Contrast	Δ	95% CI	p_{Holm}
wF1 (official)	M1-M0	-0.001	[-0.006, 0.003]	1.000
	M3-M2	+0.001	[-0.004, 0.006]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.008	[-0.018, 0.002]	0.416
	M6-M5	-0.008	[-0.016, -0.001]	0.171
Tuple accuracy	M1-M0	+0.035	[0.028, 0.043]	**0.001**
	M3-M2	+0.004	[-0.001, 0.008]	0.370
	M4-M1	-0.006	[-0.010, -0.002]	**0.028**
	M6-M3	-0.005	[-0.015, 0.004]	0.505
	M6-M5	-0.004	[-0.011, 0.004]	0.505

only confirmatory family applies Holm correction to these five contrasts on Chinese official wF1 [11]. Pre-specified Chinese tuple exact-match is a separate secondary family of five. Bracketed 95% intervals are uncorrected bootstrap percentile intervals; decisions use $p_{\text{Holm}} < .05$, not whether an interval excludes zero.

External replication was pre-registered around within-arm contrast directions; we additionally apply a report-cluster bootstrap only to the fixed English M1–M0 tuple contrast. We do not pool languages into a post-hoc significance test. C-wF1 and hF were added after the Chinese primary result, while structural-loss and alternate-backbone screens test robustness; all are exploratory and do not support confirmatory claims regardless of their nominal p-values.

6 RESULTS

Guaranteed validity and Chinese confirmatory evidence. Table 2 compares decision rules on identical base probabilities and Test rows. Independent argmax (M0) produces invalid tuples for 12.55% of document-disjoint predictions; projection and 17-state decoding (M1–M6) reduce this rate to zero by construction. This zero rate is a deterministic property of the validity layer, not evidence that the base model became more accurate. Pooled over both protocols and all three seeds, M0 emits 1,527 invalid tuples in 12,000 rows, of which 1,454 (95.2%) break one of two implications: a substantive child under PS=No, or an evidence quality under ES=No. Table 5(a) gives the full breakdown, and Table 5(b) derives the 44.3% weighted field credit those rows retain. The failure concentrates where a parent is denied but its children are answered anyway. The ± values are sample standard deviations across three seeds, each varying both folds and training, so they describe pipeline spread, not a confidence interval.

Table 3 reports both pre-specified families in full. All five official wF1 contrasts are undetected after Holm correction. For the central M1–M0 contrast, $\Delta = -0.001$ with 95% interval $[-0.006, 0.003]$ and $p_{\text{Holm}} = 1.000$. This is the absence of a detected difference; it does not establish a zero effect.

The null is not confined to that contrast. The seven rules span 0.008 in official wF1, and that spread is itself one contrast: M5 scores highest and M6 lowest, and $|\text{M6} - \text{M5}|$ has an interval excluding zero ($[-0.016, -0.001]$) yet does not survive correction ($p_{\text{Holm}} = .171$). Even the largest gap between rules is thus below what this evaluation can resolve.

Whole-tuple effects. Exact whole-tuple accuracy tells a different story. M1–M0 is +0.035 $[0.028, 0.043]$, $p_{\text{Holm}} = .001$. Conversely, 17-state decoding is not uniformly better: M4–M1 is -0.006 $[-0.010, -0.002]$, $p_{\text{Holm}} = .028$. Searching every legal state can therefore reduce whole-row correctness relative to greedy projection. The validity guarantee is shared, but the choice among legal tuples still matters.

Neither calibration contrast is detected (M3–M2 and M6–M5, on both pre-specified metrics): fitting biases only on the subset each parent admits does not improve on global biases. The calibrated projection arms still reach 0.428 and 0.432 tuple accuracy against M1's 0.394, a step no pre-specified contrast covers.

The post-hoc C-wF1 and hF M1–M0 point estimates are also positive (+0.004 and +0.003) but do not alter the inferential hierarchy; under same-document evaluation C-wF1 stays positive at +0.002, on an interval $[-0.001, 0.005]$ that crosses zero.

Evaluation protocol. Table 6 compares the two protocols, which target distinct estimands rather than a biased and an unbiased version of one quantity. Every rule scores higher under same-document evaluation, by 0.010 to 0.020, with every interval excluding zero. The gap belongs to the split, not to any rule, and exceeds the 0.008 separating the rules: which paragraphs are held out matters more than which decision rule is applied.

Multilingual replication. Table 4 fixes one arm per language before inspecting scores, each under document-disjoint evaluation at $\lambda = 0$. Whole-tuple accuracy rises in all five corpora, whereas weighted macro-F1 rises only for English and French; Section 7.1 reads its two ledger columns.

Table 4: Projection (*M*1) versus independent argmax (*M*0) on five corpora under document-disjoint evaluation; values are three-seed means for preselected backbones.

Corpus	Rows	M0 invalid %	*N/A* net	Subst. net	Δ wF1	Δ tuple acc.
Chinese (AI CUP)	2,000	12.55	+395	-264	-0.0011	**+0.0350**
English	400	23.17	+148	-86	+0.0032	**+0.0725**
French	400	21.08	+150	-107	+0.0036	**+0.0625**
Japanese	400	31.25	+183	-245	-0.0131	**+0.0708**
Korean	500	22.73	+103	-148	-0.0192	**+0.0293**

Table 5: 1,527 **invalid tuples from independent argmax across both protocols and three seeds, partitioned by violated implication and retained field credit.**

(a) Which implication the invalid tuple breaks

Violated implication	Rows	Share
PS=No, some child not *N/A*	699	45.8%
ES=No, EQ substantive	755	49.4%
PS=Yes, VT or ES *N/A*	47	3.1%
ES=Yes, EQ=*N/A*	26	1.7%
All invalid tuples	1,527	100.0%

(b) Weighted field credit those rows still collect

Field	Weight × accuracy	Credit
PS	0.20 × 0.672	0.134
VT	0.15 × 0.320	0.048
ES	0.30 × 0.360	0.108
EQ	0.35 × 0.436	0.153
Total	1.00	0.443

Table 6: Chinese anchor under same-document and document-disjoint evaluation; Δ compares the resulting estimands, with independently rounded values.

Method	Same-document	Document-disjoint	Δ	95% CI
M0	0.585	0.572	0.012	[0.004, 0.022]
M1	0.581	0.571	0.010	[0.001, 0.019]
M2	0.587	0.574	0.012	[0.001, 0.024]
M3	0.588	0.575	0.012	[0.002, 0.023]
M4	0.586	0.571	0.014	[0.005, 0.024]
M5	0.590	0.576	0.015	[0.004, 0.027]
M6	0.587	0.567	0.020	[0.011, 0.030]

Table 7 reports all four external corpora at both loss settings, arm by arm. M1–M0 tuple accuracy is positive in 32/32 arms. The weighted macro-F1 contrast is positive in only 14/32: English 7/8, French 5/8, Japanese 0/8, and Korean 2/8. These are within-arm directions, not evidence about language difficulty: the corpora also differ in source documents, label distributions, input units, and backbones, and macro-F1 weights each Korean evidence-quality class more heavily (Section 5.2). Projection thus combines a deterministic validity guarantee with a repeated complete-tuple improvement, while its field-score effect is not directionally stable across arms.

For the pre-registered fixed English arm, the one-sided tuple effect is +0.0725 $[0.0452, 0.1019]$, $p = .0002$; with only nine report clusters, this p marks a direction that survived resampling, not a precise tail probability.

Exploratory robustness. On the Chinese anchor, structural loss lowers M0's invalid-tuple rate from 12.55% to 5.18%, with a pre-registered M1 wF1 effect of +0.00250 $[-0.00384, 0.00860]$ ($p = .427$); it lowers that rate in all 16 external language-by-backbone pairs, with mixed wF1 and tuple effects. The Chinese DeBERTa and ELECTRA screens also lack conclusive wF1 gains, and the same-family base checkpoint's lower invalid rate contradicts a simple smaller-model mechanism.

Table 8 reports these seven Chinese arms. The $\lambda = 0.3$ arms still emit invalid tuples: the penalty pulls probability mass toward $\mathcal{S}$ but, unlike a decision rule, does not confine output to it. M1–M0 raises tuple accuracy in 7/7 arms and M4–M1 lowers it in 7/7, so the single-arm M4–M1 result reflects the whole screen.

7 DISCUSSION

7.1 Why the two metrics disagree

Projection's contribution is a guarantee, not a universal field-score gain, and the guarantee has a cost in some scoring views: Table 4 counts the correct field predictions projection creates and destroys, and the *N/A* net is positive while the substantive-class net is negative in all five fixed arms. If an independently predicted parent is wrong, projection may replace a correct substantive child with the structurally required *N/A*; when the parent is correct, the same operation can repair an unsupported child. Field-wise macro-F1 values these changes class by class, whereas tuple accuracy credits only four-field agreement. The field-score effect therefore varies across backbones and corpora.

Why greedy projection beats joint decoding. M4 searches all 17 legal states yet loses tuple accuracy to M1. It ranks states by the product of field probabilities, which finds the most probable row only if the fields are independent given the text, the assumption M0 makes. Pooled over protocols and seeds, it revises promise status relative to projection on 339 of 12,000 rows: 225 become correct and 114 wrong, yet whole-row correctness on those rows falls from 114 to 47. Repairing a parent rarely rescues a row whose children remain wrong, whereas breaking a correct parent always loses it. Table 8 shows the same split: without structural training, joint decoding leaves wF1 level or higher but tuple accuracy level or lower. A wider legal-state search is thus not automatically better, though it need not fail on other models or tasks.

Table 7: All 32 external arms: three-seed mean $M1 - M0$ contrasts under document-disjoint evaluation; bold denotes positive tuple-accuracy contrasts.

		$\lambda = 0$			$\lambda = 0.3$		
Corpus	Backbone	M0 inv. %	ΔwF1	Δtuple	M0 inv. %	ΔwF1	Δtuple
English	RoBERTa-large	23.17	+0.0032	**+0.0725**	13.25	+0.0033	**+0.0350**
	DeBERTa-v3-large	17.25	-0.0044	**+0.0350**	8.92	+0.0029	**+0.0175**
	ELECTRA-large	17.08	+0.0030	**+0.0400**	7.50	+0.0044	**+0.0233**
	RoBERTa-base	20.42	+0.0001	**+0.0500**	9.33	+0.0012	**+0.0217**
French	XLM-R-large	21.08	+0.0036	**+0.0625**	9.58	+0.0025	**+0.0300**
	RemBERT	21.67	-0.0117	**+0.0442**	8.92	-0.0021	**+0.0183**
	XLM-R-base	27.25	+0.0004	**+0.0650**	12.83	-0.0054	**+0.0283**
	mBERT	21.50	+0.0025	**+0.0483**	9.50	+0.0024	**+0.0150**
Japanese	XLM-R-large	31.25	-0.0131	**+0.0708**	8.58	-0.0077	**+0.0108**
	RemBERT	40.67	-0.0171	**+0.0625**	13.58	-0.0042	**+0.0142**
	XLM-R-base	37.08	-0.0043	**+0.0733**	10.67	-0.0100	**+0.0125**
	mBERT	34.67	-0.0196	**+0.0733**	8.75	-0.0060	**+0.0175**
Korean	XLM-R-large	22.73	-0.0192	**+0.0293**	8.87	-0.0032	**+0.0147**
	RemBERT	32.07	-0.0164	**+0.0513**	10.27	+0.0015	**+0.0207**
	XLM-R-base	28.93	-0.0317	**+0.0367**	10.07	-0.0047	**+0.0127**
	mBERT	21.80	-0.0245	**+0.0220**	8.07	+0.0008	**+0.0140**

Table 8: Exploratory paired Chinese backbone and structural-loss contrasts; bold denotes uncorrected 95% bootstrap intervals excluding zero.

			Enforce legality (M1−M0)		Joint decoding (M4−M1)	
Backbone	λ	M0 inv. %	wF1	Tuple acc.	wF1	Tuple acc.
Chinese RoBERTa-large	0	12.55	-0.001	**+0.035**	+0.000	**-0.006**
Chinese DeBERTa-v2-320M	0	19.75	**-0.008**	**+0.050**	**+0.013**	-0.003
Chinese ELECTRA-large	0	10.20	**-0.005**	**+0.024**	**+0.007**	-0.000
Chinese RoBERTa-base	0	11.77	**-0.007**	**+0.030**	**+0.008**	-0.003
Chinese RoBERTa-large	0.3	5.18	-0.001	**+0.014**	-0.001	-0.001
Chinese DeBERTa-v2-320M	0.3	6.53	+0.001	**+0.015**	-0.002	**-0.003**
Chinese ELECTRA-large	0.3	3.68	-0.000	**+0.009**	-0.000	-0.002

Calibration and factorization. Calibration interacts with the output rule: conditional biases do not detectably change projection (M3−M2, +0.001 wF1) but lower 17-state decoding by 0.008 (M6−M5, not surviving correction), plausibly because the pinned *N/A* biases of Eq. (4) distort the cross-state comparison only the decoder makes. We hold the factorized four-head model fixed by design: classifier chains or a 17-way classifier would change the encoder rather than the decision rule, and the latter is ill-posed here, as seven of the 17 legal states occur at most 26 times in the labeled release, two never.

7.2 Validity in audit use

Validity is a precondition for using a prediction as an audit record, not evidence that the record is right. A contradictory tuple, such as an evidence quality on a paragraph judged to have no evidence, cannot be filed or counted without deciding which field to believe; projection decides deterministically, making every output admissible but not correct. A repair can turn a visible inconsistency into a consistent error, removing the cue that would have drawn an auditor's attention, so a deployed validity layer should log which rows it repaired.

Choosing between the metrics. The metrics disagree for two separable reasons: wF1 credits each correct field even in a wrong or contradictory row and weights rare classes like common ones, whereas tuple accuracy credits only whole rows, weighted by frequency. The choice should follow how outputs are used. (i) If each paragraph becomes one record, such as an assurance file or a database entry, one wrong field spoils it; compare rules by tuple accuracy among valid outputs, which here favors projection over joint decoding. (ii) If one field drives an action, such as routing promises to analysts, use that field's own F1, and for a rare target such as misleading evidence its per-class recall. (iii) If outputs feed portfolio statistics, such as the share of promises with clear evidence, use class-level field scores but keep validity, since invalid tuples make such counts inconsistent. (iv) Use official wF1 only where it is the agreed rule, as on a leaderboard, and not to separate rules closer than its resolution here (0.008).

7.3 Reproducibility

The repository releases row-level predictions and text-free metadata for all eight paper tables and Figure 1; raw corpora, model weights, and retraining are excluded. Configurations pin each backbone's Hugging Face revision. Per seed, reports are identifier-sorted, shuffled, stable-sorted by decreasing size, and each assigned uniformly to one of the two lightest folds, ties broken by fold index. Korean text is one page per labeled (URL, page) pair, unique in the release, from Poppler's `pdftotext -f P -l P -layout` or, for image-only pages, `pdftoppm -r 250` and Tesseract (`-l kor+eng`). The 32 external arms took 25.4 GPU-hours on one RTX 3090.

7.4 Limitations

The replication matrix is not a factorial estimate of a pure language effect. English and French have only nine report clusters each and Chinese 49, so the 32/32 and 14/32 summaries are descriptive directions across executed arms, not a pooled significance test. AI CUP weights applied to ML-Promise are not an official ML-Promise metric. Every contrast uses M0, unconstrained argmax on the same probabilities, to isolate the decision rule. We do not compare against a published system, official baseline, or leaderboard and make no participant-ranking claim.

Korean page inputs (Section 3) are not redistributed: released predictions can be rescored, but retraining requires the recipe in Section 7.3.

Structural-loss and backbone results are exploratory; fixed loss weights and few checkpoints cannot characterize all objectives or architectures. The rarest Chinese quality label has two examples and Korean none, so no class-level claim is made. Each seed changes folds and training randomness together, so their effects cannot be separated.

8 CONCLUSION

Hierarchical projection is not a guaranteed route to higher field-wise weighted macro-F1. It is instead a retraining-free validity layer that eliminates invalid tuples by construction and improves exact whole-tuple accuracy on the Chinese anchor and in all 32 evaluated external arms. Its field-score effect varies across models and corpora, but its logical guarantee does not. For audit use, validity should be a precondition, and the metric that picks a rule should match the unit at which errors have consequences.

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